

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals and Data Structures	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated: 19th March, 2026
Semester: 1st	Session: 2026
Lab/Project/Assignment #: 6	CLOs to be covered: CLO2
Lab Title: Local and Global Variables	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO2	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Mark s	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 6: Local and Global Variables

Lab Goals/Objectives:

To understand the difference between local and global variable scope in Python, and how the globals() function can be used to access global variables from within a function.

Equipment Required:

Python 3, VS Code / PyCharm

Theory:

A variable defined inside a function is local to that function and is not accessible outside it. A variable defined outside all functions is global and can be accessed anywhere in the program. When a local variable has the same name as a global variable, the local variable takes precedence inside the function, but the global one can still be accessed using `globals()`.

Lab Work:**Task 1: Local vs Global Variable Scope**

```
x=45
def a():
    x=40
    print(x)
    print(globals()["x"])
a()
```

Result:

```
40
45
```

Conclusions:

The concept of local versus global variable scope was demonstrated, including how a local variable shadows a global variable of the same name inside a function.